

THEORY AND METHODS

SMALL SAMPLE PERFORMANCE OF EXTREME REGRESSION QUANTILES

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Key words: Bounded influence; heavy tails; influential points; regression depth; restricted estimators; skew distributions; symmetric distributions; trimmed means.

Summary: In this paper we studied the finite sample behaviour of a range of estimators for extreme regression quantiles. We studied the behaviour of the estimators over a wide range of heavy tailed error distributions and design matrices containing influential design points. Our main conclusion is that restricted bounded influence estimators should be the estimators of choice. In particular the preferred estimator is a restricted bounded influence Koenker-Bassett based on a common slope estimator of a 20% trimmed mean. This estimator has good performance in well-behaved cases and protects one against the adverse effects of heavy tails and high leverage design points.

¹ The research was partially supported by National Research Foundation grant 2053331 and a research grant from Potchefstroom University for CHE.

² The research was partially supported by National Research Foundation grant 2046922 and a research grant from the University of Stellenbosch.

1. Introduction

Since the introduction of regression quantiles (RQ) by Koenker and Bassett in 1978, they have been studied and applied extensively. However, most of the work was done for “central” RQs. Early exceptions were papers by Portnoy (1987), by Bassett (1988) and more recently by Smith (1994), where the “maximum” and “minimum” RQs were studied. In the last 2 years there has been an increasing interest in these extreme RQs, see e.g. Portnoy and Jureckova (2000), Jureckova (2000), Chernozhukov (2000). Results on the theoretical properties of extreme RQs and functions of them were derived and a number of examples given where extreme RQs play a role in applications. However, these results are mostly of an asymptotic type and not much is known about their behaviour in small samples. We focus on such results in this paper.

Recently Rousseeuw and Hubert (1999) introduced a “new” notion of regression depth (RD) and the concomitant maximum regression depth estimators (MRDs). These estimators have very favourable robustness properties, in both the independent variable space and the covariate space. The concept was generalised in a natural fashion to regression depth quantiles (RDQs) following a suggestion by Koenker on an early version of the paper by Rousseeuw and Hubert. Some results on the asymptotics of such MRD estimators and RDQs were obtained by He and Portnoy (2000). Again these results were of an asymptotic type while our interest is in their small sample behaviour.

The aim of this paper was to study the numerical behaviour of extreme RQs and extreme RDQs in finite samples in order to see how well they estimated the corresponding population quantities and how robust they were to longer

tailed distributions as well as to ill-behaved design matrices. We carried out a simulation study for a range of different estimators, over a range of error distributions and design matrices.

In section 2 we define the estimators considered in more detail and in section 3 we discuss the simulation study design. The simulation results and discussion are given in section 4 and some conclusions in the last section.

2. Definitions of estimators

Consider the usual linear model $Y_i = \underline{x}_i' \underline{\beta} + e_i$, $i = 1, 2, \dots, n$, with $\{e_i\}$ iid errors having common distribution function F and $X = \begin{pmatrix} \underline{x}_1' \\ \vdots \\ \underline{x}_n' \end{pmatrix}$ an $n \times p$ fixed design matrix containing an intercept term. Here $\underline{\beta}$ is a $p \times 1$ vector of unknown regression parameters.

For any $\underline{b} \in R^p$, let

$$r_i \equiv r_i(\underline{b}) = Y_i - \underline{x}_i' \underline{b}, \quad i = 1, \dots, n$$

denote the corresponding residuals.

2.1 Koenker-Bassett RQs

For $0 < \theta < 1$, the θ -th RQ is given by

$$\begin{aligned} \widehat{\underline{\beta}}_{RQ}(\theta) &= \arg \min_{\underline{b}} \sum_{i=1}^n |r_i| [\theta I(r_i \geq 0) + (1 - \theta) I(r_i < 0)] \\ &= \arg \min_{\underline{b}} \sum_{i=1}^n r_i [\theta - I(r_i < 0)], \end{aligned}$$

with $I(\cdot)$ denoting the indicator function.

Since the $\underline{x}' \widehat{\underline{\beta}}_{RQ}(\theta)$ are not necessarily monotone increasing in θ at all $\underline{x}$, so-called restricted RQs (RRQs) were introduced by Koenker (1984), He (1997), Zhao (2000) as follows.

Write $\underline{\beta}' = (\beta_0, \underline{\beta}'_1)$ with β_0 the intercept parameter. Estimate $\underline{\beta}_1$, the vector of slope parameters, by e.g. (i.e. $\theta = \frac{1}{2}$ in the above), say $\hat{\underline{\beta}}_1$. This common slope guarantees that all quantile planes are parallel with no crossings. Then estimate the intercept at different quantiles by the sample quantile $\hat{\beta}_0(\theta)$ of the residuals obtained from the first step, viz. $\hat{r}_i = Y_i - \underline{x}'_i \hat{\underline{\beta}}_1$, $i = 1, 2, \dots, n$, where $\underline{x}'_i = (1 \underline{x}'_{1i})$. The θ -th RRQ estimator is then given by $(\hat{\beta}_0(\theta), \hat{\underline{\beta}}'_1)$.

In order to guard against outliers in the design space, we also consider bounded influence (BI) RQs and BI-RRQs. BI-RQs are defined as follows. For weights $\{w_i\}$ let $\hat{\underline{\beta}}_{RQ}^{(w)}(\theta) = \arg \min_{\underline{b}} \sum_{i=1}^n w_i r_i [\theta - I(r_i < 0)]$. We will use the Mallows weights (see e.g. Mallows, 1973 or de Jongh, de Wet and Welsh, 1988). Combining restricted and bounded influence quantiles, we define bounded influence restricted regression quantiles (BI-RRQs) in an obvious fashion.

2.2 Rousseeuw – Hubert RDQs

The maximum regression depth (MRD) estimator is defined as the $\underline{b}$ having the largest regression depth, i.e.

$$\hat{\underline{b}}'_{MRD} = \arg \max_{\underline{b}} [rdepth(\underline{b}, data)],$$

where

$$rdepth(\underline{b}, data) = \min_V [\min (L^+(V) + R^-(V), R^+(V) + L^-(V))]$$

with V any affine hyperplane, which splits the data into 2 sets say $L(V)$ and $R(V)$, and $L^+(V)$ ($L^-(V)$) denoting the number of residuals in $L(V)$ with positive (negative) residuals with respect to the hyperplane $\underline{x}'\underline{b}$. Similarly for $R(V)$. See Rousseeuw and Hubert (1999) for these definitions.

In an analogous fashion, let the θ -th Regression Depth Quantile (RDQ) be defined as

$$\hat{\beta}_{RDQ}(\theta) = \arg \max_{\underline{b}} [\theta - rdepth(\underline{b}, data)] ,$$

where

$$\begin{aligned} & \theta - rdepth(\underline{b}, data) \\ &= \min_V [\min \{ \theta L^+(V) + (1 - \theta) R^-(V), \theta R^+(V) + (1 - \theta) L^-(V) \}] . \end{aligned}$$

These were suggested by Koenker and discussed in Rousseeuw and Hubert (1999). One can again force these quantile hyperplanes to be parallel by defining restricted RDQs (RRDQs) as before. As common slope we use the estimator obtained from the deepest regression fit.

Remarks

- (i) Note that all restricted estimators are based on a common slope while the unrestricted estimators are not.
- (ii) Non-unique solutions of the depth and Koenker-Bassett regression quantiles may exist. Here unique means that there is a uniquely determined intercept and slope estimate that give the quantile solution. In the case of the Koenker-Bassett quantiles the (slightly adapted) Barrodale-Roberts linear programming algorithm provides the uniqueness indicator as standard output (see e.g. de Jongh, 1985). The solution provided when non-unique solutions exist is arbitrarily chosen from the set of solutions. Similarly the non-uniqueness of the depth quantiles is easily determined (see Rousseeuw and Hubert, 1999). Note that if a non-unique solution exists, there might exist more than two possible solutions. In the case when non-unique solutions exist, the depth quantile is determined as an average of the possible solutions (see Rousseeuw

and Hubert, 1999).

3. Simulation study design

The factors considered in the simulation were the sample size, the number of parameters, the error distributions, the design matrices, the estimators and the quantile choices. Each of these factors will be discussed under a separate heading below.

We generated data according to the model

$$y_i = x_i + \varepsilon_i; \quad i = 1, \dots, n$$

where y_i is the i -th observation, x_i the i -th design point and ε_i the i -th error, distributed according to some distribution F and n the sample size. Without loss of generality we assumed a zero intercept term and slope parameter equalling one.

3.1 Sample size

We considered two sample sizes, viz. $n = 21$ and $n = 41$.

3.2 Number of parameters

We considered only the case of 2 regression parameters, i.e. $p = 2$. The reason for this restriction was the computational complexity of the RD estimators at higher p values.

3.3 Error distributions

Let $CN(a, b)$ denote a contaminated normal distribution, viz. $(1 - a)N(0, 1) + aN(0, b^2)$. We considered the contaminated normals to represent symmetric, heavier tailed (than normal) distributions. Our study also included the t-distribution with 3 degrees of freedom. In order to also consider non-symmetric distributions, we included a number of gamma distributions.

In the table below we list the error distributions considered together with some of their properties in order to illustrate their degree of heavy tailedness and non-symmetry.

Table 1. Distributional properties

	Distribution	Variance	Kurtosis	Skewness
1	$N(0, 1)$	1	3	0
2	$CN(.1, 3)$	1.8	8.3	0
3	$CN(.2, 3)$	2.6	7.5	0
4	$CN(.1, 10)$	10.9	25.3	0
5	t_3	3	—	0
6	$ga\left(2, (0.9)^{\frac{1}{2}}\right)$	1.8	6	1.4
7	$ga\left(2, (5.45)^{\frac{1}{2}}\right)$	10.9	6	1.4
8	$ga\left(1, (1.8)^{\frac{1}{2}}\right)$	1.8	9	2
9	$ga\left(1, (10.9)^{\frac{1}{2}}\right)$	10.9	9	2
10	$ga\left(.5, (3.6)^{\frac{1}{2}}\right)$	1.8	15	2.8
11	$ga\left(.5, (21.8)^{\frac{1}{2}}\right)$	10.9	15	2.8

3.4 Design matrices

Five design matrices were used in the study, including a well-behaved one and four with varying degrees of ill-behaviour. Note that each design matrix has a first column of 1's.

- M1: well behaved
 $x_i = i, \quad i = 1, 2, \dots, n$
- M2: one influential observation in a positive direction as M1 with $x_{(n+1)/2}$ replaced by 61
- M3: two influential observations in a positive direction as M1 with $x_{(n+1)/2}$ replaced by 61, x_n replaced by 71
- M4: two influential observations in opposite directions as M1 with x_1 replaced by -49, x_n replaced by 61
- M5: one large influential observation in a positive direction as M1 with $x_{(n+1)/2}$ replaced by 300

3.5 Estimators

The estimators studied and their abbreviations used, are listed below.

KB:	Koenker-Bassett regression quantiles
BIKB:	Bounded influence K-B regression quantiles
DEPTH:	Regression depth quantiles
RRKB:	Restricted K-B regression quantiles using L_1 as common slope estimator
RRDEPTH:	Restricted depth quantiles using deepest regression as common slope estimator
RRBITM:	Restricted bounded influence K-B regression quantiles using 20% regression trimmed mean as common slope estimator
RRBIKB:	Restricted bounded influence K-B using L_1 as common slope estimator

Note that the 20% regression trimmed mean is the least squares estimator of the observations remaining after discarding observations lying outside the 10% and 90% Koenker-Bassett regression quantiles.

3.6 Quantile choices

Since our interest was mostly in the extremes, we concentrated on the range $\theta = 0.025, 0.05, 0.01$ and symmetrically $\theta = 0.975, 0.95, 0.90$ for $n = 41$ and $\theta = 0.025, 0.05$ and again symmetrically $\theta = 0.975, 0.95$ for $n = 21$. In tables 2 and 3 we give the results for wider ranges of θ , since these other choices may also be of interest to our readers. Note that for the restricted estimators, the smallest order statistic corresponds to approximately $\theta = \frac{1}{n}$, giving $\theta = 0.025$ for $n = 41$ and $\theta = 0.05$ for $n = 21$.

4. Numerical results

Our main criterion for comparing the estimators under different conditions was through MSE, and its two components of variance and bias.

Before giving these, we first discuss two other important issues, viz. uniqueness of quantile solutions and actual trimming proportion achieved.

4.1 Unique solutions

During our simulations, we calculated at each simulation run and at each quantile choice, whether the solution (in order to obtain that particular quantile) was unique or not. These we averaged over distributions as well as over design matrices. We did this since the results were very similar over design matrices and error distributions.

Our results are given in the following table (for $n = 41$ and 1000 simulations).

Table 2. Average proportion of unique quantile solutions

	$\theta = 0.025$	$\theta = 0.10$	$\theta = 0.25$	$\theta = 0.50$
BIKB	0.99	1.00	1.00	1.00
DEPTH	0.63	0.73	0.39	0.24
KB	1.00	1.00	0.98	0.95
RRBIKB	1.00	1.00	1.00	1.00
RRDEPTH	0.24	0.24	0.24	0.24
RRKB	0.95	0.95	0.95	0.95

Note that, because RRBITM is not based on a quantile fit, it is not included in the table.

Remarks

- (i) The Koenker-Bassett estimators are mostly uniquely determined, with the exception of the extreme quantiles of $\theta \leq 0.001$.
- (ii) The estimators DEPTH are clearly not uniquely determined in most cases, with the non-uniqueness most severe for the more central quantiles. McKean and Sheather in their discussion of the paper by Rousseeuw and Hubert (1999) on regression depth, concentrated on the variability of deep lines and the effect that outliers have on deep

line fits. In particular, they emphasised that deepest lines are not unique and that often many deepest lines exist. They also pointed out that “the fits of these deepest lines can vary considerably, and hence, that depending on the choice of the deepest line, quite different interpretations can occur”. In the small simulation study that they conducted, they found that the number of deepest lines varies from 1 to 16 with the median number being 3.

In our simulation study we counted the number of deepest lines, as well as the number of depth quantiles at a range of quantile fits. We found that over all design matrices and distributions considered, the average number of depth quantile lines is 3.47 for $\theta = 0.05$, 1.36 for $\theta = 0.10$ and 1.4 for $\theta = 0.025$. The respective SDs are 5.4 ($\theta = 0.05$), 2.0 ($\theta = 0.10$) and 2.1 ($\theta = 0.025$). Thus for $\theta = 0.5$, on average 3.47 deepest regression line fits were used to determine the resultant “average” deepest line fit.

Note that the average and SDs reported here are fairly representative of all the design matrices and distributions considered, i.e. they stayed fairly constant over those factors. However, there is some indication that at the more extreme quantiles ($\theta = 0.10$ and 0.025), the average number of depth quantile fits seemed to be slightly more in the presence of influential observations or outliers.

4.2 Achieved quantile proportion

We now give the achieved quantile proportion, defined as the proportion of points strictly below a quantile line. This proportion has to be compared to the corresponding target value θ . The results are again averaged over design matrices and error distributions ($n = 41$ and 1000 simulation runs).

Table 3. Achieved quantile proportion

Estimator	$\theta=0.025$	$\theta=0.05$	$\theta=0.1$	$\theta=0.2$	$\theta=0.3$	$\theta=0.4$	$\theta=0.5$
BIKB	0.000	0.0275	0.0763	0.1762	0.2763	0.3758	0.4757
DEPTH	0.0710	0.0899	0.1304	0.2182	0.2994	0.3922	0.4922
KB	0.0012	0.0257	0.0756	0.1758	0.2757	0.3758	0.4761
RRBIKB	0.0244	0.0488	0.0976	0.1951	0.2927	0.3902	0.4763
RRBITM	0.0244	0.0488	0.0976	0.1951	0.2927	0.3902	0.4878
RRDEPTH	0.0244	0.0488	0.0976	0.1951	0.2927	0.3902	0.4868
RRKB	0.0244	0.0488	0.0976	0.1951	0.2927	0.3902	0.4761

Remarks

- (i) Restricted estimators achieve the correct proportion, due to the way they are constructed.
- (ii) KB and BIKB tend to achieve quantile proportions lower than the target values. Note that they each have 2 points on the line, which are not counted in the numerator when calculating the quantile proportions.
- (iii) DEPTH tends to achieve proportions higher than the target values at the lower quantiles.

4.3 MSE results

As mentioned in the introduction the purpose of this study was to determine the small sample behaviour of extreme regression quantiles. We were interested in the behaviour of these under both heavy tailed error distributions and influential design points. Our primary measure of performance of the estimators was their mean squared error (MSE) and we report on these in this subsection. In order not to clutter the body of the paper, we will give most of the tables in an appendix while concentrating here on summarised results. We only give the results for $n = 41$; results for $n = 21$ are similar and available upon request.

We summarised our results in two respects

- calculating efficiencies over estimators, and
- graphing results summarised over distributions.

We now give and discuss each of these in turn.

4.3.1 Efficiencies

For each design matrix and each error distribution, we calculated the efficiency of a particular estimator with respect to the best estimator in the column, i.e. for estimator A its efficiency $eff(A)$ is defined as

$$eff(A) = \frac{\min_{\text{all estimators}} MSE}{MSE(A)} \times 100.$$

These results are given in tables 4 to 8 below, for the 5 design matrices M1 to M5 respectively. Note that the distributions are numbered as in table 1, so that distributions 1–5 are symmetric and distributions 6–11 are skew.

A discussion of these tables follows.

Design M1 is a well-behaved design and we find that the restricted estimators do the best over all the symmetric distributions, except at the distribution $CN(\cdot, 1, 10)$ where there is some loss of efficiency at $\theta = 0.05, 0.95$. In these cases it is outperformed by the estimator DEPTH. Bear in mind that $CN(\cdot, 1, 10)$ is an extremely heavy tailed distribution.

For the non-symmetric gamma distributions all the restricted estimators perform very well at the upper quantiles (long tail). However, at the lower quantiles, there is a consistent loss of efficiency of the restricted estimators relative to especially the KB estimator. This loss of efficiency is more pronounced towards more extreme quantiles. Note, however, that for these distributions the lower tails are not of particular interest.

In summary, in the case of heavy tailed distributions and a well-behaved design, the restricted estimators perform very well in most cases of interest. Exceptions are the high kurtosis gamma distributions at the low quantiles (i.e. the short tail).

Design M2 has a high influential point placed in such a way that it adversely influences estimation of the lower quantiles. For the higher quantiles the conclusions are thus similar to that for M1. For the lower quantiles, and for the symmetric distributions, the restricted estimators again perform very well. Also, for the gamma distributions their performance is quite respectable.

The results for design M5 are roughly similar to those for M2. This is to be expected since M5 is simply a more severe M2. Note that now the restricted estimators that do not have the bounded influence property, have a substantial loss in efficiencies.

Design M3 has 2 influential points on the same side which blow up the MSE of all estimators at the most extreme quantiles due to severe bias. This is to be expected. This negatively influences even a generally good estimator such as RRBITM. However, as soon as one moves above the most extreme quantile, the MSEs of the restricted estimators show dramatic improvement. The effect of the high leverage points is now neutralised.

We note that, except for the most extreme lower quantile (.025) the restricted, bounded influence estimators do well at all symmetric distributions and at all distributions at the upper quantiles. At the lower quantiles for the gamma distributions (their short tail) these estimators have some (variable) loss of efficiency. Although this is the case, the absolute sizes of all the MSEs are very small.

Table 4. Efficiencies under design matrix M1

		Distributions									
		1	2	3	4	5	6	7	8	9	10
Q=0.025	BIKB	99%	42%	79%	67%	20%	99%	96%	100%	95%	98%
	DEPTH	33%	61%	52%	100%	39%	70%	67%	37%	35%	99%
	KB	100%	44%	80%	73%	20%	100%	100%	98%	100%	99%
	RFBIKB	98%	99%	100%	99%	99%	57%	71%	72%	36%	39%
	RFBITM	98%	96%	90%	99%	99%	68%	61%	79%	85%	36%
	RFDEPTH	100%	98%	100%	99%	99%	60%	52%	69%	76%	42%
	RFKB	100%	100%	99%	99%	100%	58%	55%	70%	77%	39%
Q=0.05	BIKB	92%	52%	60%	29%	99%	99%	95%	100%	95%	98%
	DEPTH	39%	68%	70%	100%	52%	74%	72%	49%	46%	99%
	KB	97%	50%	62%	27%	59%	100%	100%	99%	100%	98%
	RFBIKB	98%	100%	100%	60%	100%	68%	66%	82%	82%	47%
	RFBITM	98%	99%	97%	60%	100%	80%	77%	91%	96%	50%
	RFDEPTH	100%	100%	99%	60%	99%	70%	66%	80%	87%	49%
	RFKB	99%	100%	99%	60%	99%	68%	69%	81%	89%	51%
Q=0.10	BIKB	98%	70%	71%	19%	83%	98%	96%	100%	94%	98%
	DEPTH	52%	57%	64%	73%	64%	73%	75%	58%	56%	99%
	KB	100%	71%	73%	21%	81%	100%	100%	98%	100%	98%
	RFBIKB	97%	97%	98%	100%	97%	87%	81%	82%	88%	70%
	RFBITM	97%	100%	100%	100%	100%	98%	97%	91%	99%	75%
	RFDEPTH	98%	99%	97%	100%	98%	88%	83%	83%	90%	71%
	RFKB	97%	98%	99%	98%	100%	86%	86%	81%	90%	74%
Q=0.90	BIKB	99%	81%	62%	31%	70%	88%	94%	92%	90%	91%
	DEPTH	50%	53%	63%	100%	58%	60%	59%	57%	53%	68%
	KB	100%	79%	64%	31%	79%	91%	96%	94%	93%	92%
	RFBIKB	96%	97%	97%	90%	100%	100%	96%	96%	96%	99%
	RFBITM	97%	100%	100%	93%	100%	100%	100%	98%	100%	100%
	RFDEPTH	96%	100%	98%	91%	99%	100%	99%	99%	99%	99%
	RFKB	93%	96%	96%	92%	100%	100%	96%	100%	100%	96%
Q=0.95	BIKB	93%	54%	71%	31%	67%	79%	74%	90%	80%	82%
	DEPTH	40%	64%	77%	100%	59%	48%	44%	49%	46%	51%
	KB	98%	55%	73%	28%	57%	78%	81%	96%	88%	87%
	RFBIKB	96%	100%	99%	99%	100%	99%	100%	100%	99%	100%
	RFBITM	97%	100%	99%	60%	100%	99%	99%	99%	99%	99%
	RFDEPTH	100%	100%	100%	59%	100%	100%	100%	99%	100%	100%
	RFKB	97%	100%	100%	60%	99%	99%	100%	100%	100%	99%
Q=0.975	BIKB	94%	51%	83%	68%	26%	80%	76%	84%	79%	79%
	DEPTH	34%	63%	54%	95%	46%	42%	37%	40%	41%	45%
	KB	97%	51%	88%	70%	30%	82%	78%	85%	79%	79%
	RFBIKB	98%	99%	100%	100%	100%	99%	99%	100%	99%	100%
	RFBITM	99%	100%	100%	100%	99%	99%	99%	99%	99%	99%
	RFDEPTH	100%	100%	99%	100%	100%	100%	100%	99%	100%	100%
	RFKB	98%	99%	100%	100%	99%	100%	100%	100%	100%	99%

Table 7. Efficiencies under design matrix M4

		Distributions										
		1	2	3	4	5	6	7	8	9	10	11
Q=0.025	BMB	1%	16%	21%	17%	33%	0%	4%	1%	5%	0%	3%
	DEPTH	1%	19%	28%	100%	48%	0%	4%	1%	5%	0%	3%
	KB	1%	16%	20%	17%	33%	0%	4%	1%	5%	0%	3%
	AFBKS	97%	100%	100%	16%	100%	91%	97%	93%	88%	93%	97%
	AFBIM	100%	100%	100%	16%	100%	100%	100%	100%	100%	92%	99%
	AFDEPTH	98%	99%	100%	16%	100%	88%	92%	94%	87%	100%	100%
	ARKB	79%	96%	97%	16%	99%	33%	37%	36%	36%	36%	36%
Q=0.05	BMB	70%	50%	70%	49%	64%	100%	100%	100%	100%	100%	100%
	DEPTH	70%	59%	86%	100%	80%	94%	91%	88%	93%	94%	94%
	KB	1%	4%	7%	25%	8%	0%	2%	0%	3%	0%	1%
	AFBKS	93%	98%	100%	40%	97%	48%	50%	74%	69%	34%	36%
	AFBIM	100%	100%	99%	40%	100%	57%	54%	79%	82%	31%	36%
	AFDEPTH	93%	97%	99%	40%	97%	49%	47%	74%	71%	34%	37%
	ARKB	76%	90%	93%	41%	83%	19%	19%	29%	31%	12%	13%
Q=0.10	BMB	97%	69%	69%	36%	68%	100%	100%	100%	100%	100%	100%
	DEPTH	67%	84%	96%	100%	100%	91%	89%	84%	86%	96%	96%
	KB	9%	4%	5%	8%	5%	9%	11%	7%	12%	11%	15%
	AFBKS	96%	96%	100%	61%	98%	72%	69%	89%	79%	53%	58%
	AFBIM	100%	100%	98%	61%	100%	84%	76%	96%	91%	51%	56%
	AFDEPTH	96%	96%	98%	60%	99%	71%	66%	92%	79%	51%	57%
	ARKB	80%	73%	83%	59%	84%	31%	30%	38%	40%	18%	20%
Q=0.30	BMB	94%	76%	67%	48%	70%	79%	79%	82%	86%	89%	79%
	DEPTH	86%	92%	92%	87%	86%	84%	85%	79%	91%	100%	94%
	KB	0%	0%	0%	2%	0%	1%	7%	1%	6%	1%	9%
	AFBKS	96%	96%	98%	99%	96%	96%	96%	96%	97%	97%	99%
	AFBIM	100%	100%	100%	100%	100%	100%	100%	100%	100%	99%	100%
	AFDEPTH	96%	96%	99%	99%	96%	97%	97%	96%	97%	97%	99%
	ARKB	83%	78%	86%	97%	86%	99%	99%	96%	94%	96%	99%
Q=0.55	BMB	72%	48%	69%	79%	42%	64%	78%	58%	73%	58%	75%
	DEPTH	59%	36%	60%	100%	40%	53%	90%	50%	88%	54%	100%
	KB	0%	1%	2%	23%	2%	2%	18%	1%	13%	3%	24%
	AFBKS	97%	96%	96%	91%	96%	96%	96%	96%	96%	96%	94%
	AFBIM	100%	100%	100%	91%	100%	100%	100%	100%	100%	100%	98%
	AFDEPTH	96%	96%	100%	91%	99%	96%	96%	96%	96%	99%	94%
	ARKB	79%	95%	97%	91%	96%	100%	96%	96%	96%	99%	99%
Q=0.975	BMB	2%	15%	16%	65%	39%	15%	62%	9%	46%	21%	42%
	DEPTH	0%	4%	5%	100%	15%	6%	92%	3%	53%	10%	100%
	KB	0%	4%	5%	47%	13%	5%	46%	3%	32%	8%	36%
	AFBKS	95%	99%	99%	66%	100%	96%	96%	96%	96%	99%	99%
	AFBIM	100%	100%	100%	66%	99%	100%	99%	100%	100%	100%	94%
	AFDEPTH	96%	99%	99%	66%	100%	99%	96%	99%	99%	99%	99%
	ARKB	77%	97%	98%	66%	100%	100%	100%	99%	97%	100%	94%

		Distributions										
		1	2	3	4	5	6	7	8	9	10	11
Q=0.025	BIBK	9%	20%	17%	49%	34%	21%	28%	9%	10%	88%	86%
	DEPTH	0%	1%	1%	25%	2%	0%	0%	0%	0%	0%	0%
	KB	0%	1%	1%	14%	3%	0%	0%	0%	0%	0%	0%
	FFBIBK	93%	99%	98%	98%	99%	80%	84%	81%	80%	88%	88%
	FFBFTM	100%	100%	100%	98%	100%	100%	100%	100%	100%	91%	98%
	FFDEPTH	98%	100%	98%	98%	100%	84%	90%	84%	88%	100%	100%
	FFKB	28%	77%	77%	100%	95%	4%	4%	4%	5%	4%	4%
Q=0.05	BIBK	48%	28%	38%	46%	19%	100%	100%	84%	79%	100%	100%
	DEPTH	32%	22%	29%	46%	18%	81%	87%	58%	63%	90%	88%
	KB	0%	0%	0%	5%	0%	0%	0%	0%	0%	0%	0%
	FFBIBK	98%	98%	99%	100%	99%	65%	71%	82%	80%	43%	43%
	FFBFTM	100%	100%	99%	100%	100%	82%	88%	100%	100%	49%	48%
	FFDEPTH	97%	98%	100%	100%	99%	68%	76%	83%	84%	49%	48%
	FFKB	22%	50%	62%	99%	63%	3%	3%	4%	4%	2%	2%
Q=0.10	BIBK	58%	41%	51%	40%	45%	84%	88%	76%	71%	100%	100%
	DEPTH	80%	39%	69%	72%	55%	71%	79%	67%	66%	95%	92%
	KB	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
	FFBIBK	92%	93%	94%	98%	93%	84%	88%	88%	83%	66%	68%
	FFBFTM	100%	100%	100%	100%	100%	100%	100%	100%	100%	79%	77%
	FFDEPTH	93%	96%	97%	100%	98%	82%	87%	91%	89%	73%	73%
	FFKB	19%	20%	31%	70%	29%	4%	4%	5%	5%	3%	2%
Q=0.30	BIBK	88%	61%	48%	28%	58%	71%	71%	71%	74%	68%	67%
	DEPTH	48%	49%	82%	65%	50%	54%	52%	48%	52%	51%	58%
	KB	25%	25%	26%	16%	27%	84%	83%	74%	72%	88%	100%
	FFBIBK	96%	99%	98%	100%	96%	86%	94%	99%	98%	98%	94%
	FFBFTM	100%	100%	100%	99%	100%	96%	99%	100%	100%	98%	98%
	FFDEPTH	99%	98%	99%	99%	98%	97%	98%	98%	100%	100%	94%
	FFKB	22%	23%	26%	40%	28%	100%	100%	82%	90%	94%	100%
Q=0.95	BIBK	88%	43%	48%	27%	43%	60%	63%	65%	68%	63%	68%
	DEPTH	37%	52%	68%	100%	53%	48%	40%	38%	43%	45%	48%
	KB	33%	30%	43%	28%	30%	81%	81%	76%	73%	81%	90%
	FFBIBK	95%	97%	99%	62%	98%	100%	89%	94%	92%	100%	98%
	FFBFTM	100%	100%	100%	62%	100%	100%	89%	95%	99%	100%	100%
	FFDEPTH	98%	98%	100%	62%	99%	100%	89%	95%	99%	100%	100%
	FFKB	26%	43%	68%	60%	57%	100%	100%	100%	100%	94%	98%
Q=0.975	BIBK	77%	34%	48%	48%	24%	58%	51%	49%	61%	63%	58%
	DEPTH	31%	53%	44%	76%	48%	39%	36%	30%	33%	41%	42%
	KB	44%	34%	48%	67%	33%	71%	70%	60%	70%	80%	74%
	FFBIBK	94%	97%	81%	82%	94%	93%	91%	83%	83%	98%	98%
	FFBFTM	100%	100%	82%	82%	95%	92%	92%	84%	83%	99%	100%
	FFDEPTH	98%	100%	82%	83%	95%	93%	92%	84%	83%	100%	100%
	FFKB	36%	89%	100%	100%	100%	100%	100%	100%	100%	93%	98%

The results for design M5 are roughly similar to those for M2. This is to be expected since M5 is simply a more severe M2. Note that now the restricted estimators that do not have the bounded influence property, have a substantial loss in efficiencies.

In summary, except for the most extreme quantile, where all MSEs blow up, restricted, bounded influence estimators perform very well in all cases of interest.

Finally, design M4 has two influential points in opposite directions. Here one of these points influences the lower quantiles while the other influences the upper quantiles. For the symmetric distributions these influences are similar, as one would expect and again the restricted, bounded influence estimators handle them quite comfortably (as in the case of M2 and M5). For the gammas, we again have a variable loss of efficiency of these estimators at the lower quantiles.

As an overall conclusion as far as efficiencies are concerned, the restricted, bounded influence estimators (i.e. RRBIB, RRBIM, RRDEPTH) perform very well. The performance of these 3 estimators is very similar, with RRBIM having a slight edge over the other two.

4.3.2 Summarised graphs

In order to obtain a more compact impression of the behaviour of the estimators, we averaged the MSEs over symmetric distributions and (separately) over the gammas. These results are (selectively) shown in the graphs below, over the different quantiles, for each design matrix. For the symmetric distributions, we did not include the $CN(-1, 10)$ distribution in the averaging due to its very extreme nature.

In order to save space, for the symmetric distributions, we only give the graphs for designs M1, M2 and M3 at the lower quantiles. The graphs for M4 and M5 at the lower quantiles are very similar to that of M2. Also, with the exclusion of M4, the graphs for the upper quantiles are very similar to M1. (This is due to the absence of influential points for those designs at the upper quantiles). M4 on the other hand, has a mirror image behaviour to M2 at the lower quantiles.

Furthermore, we only included the estimators KB, DEPTH and RRBITM, these being representative of the range of estimators considered. All the graphs not shown here are available from the authors upon request.

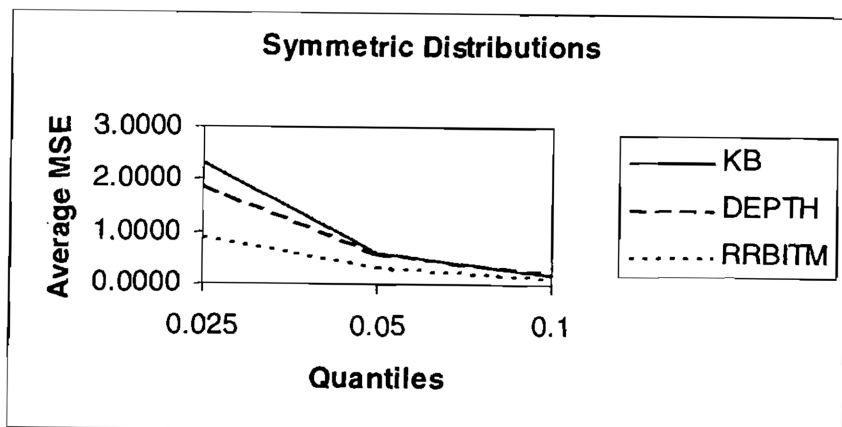


Figure 1. Average MSE for Design Matrix M1

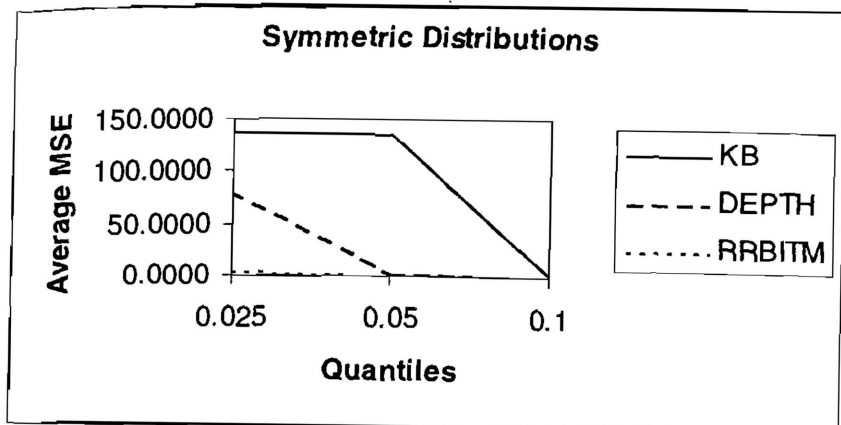


Figure 2. Average MSE for Design Matrix M2

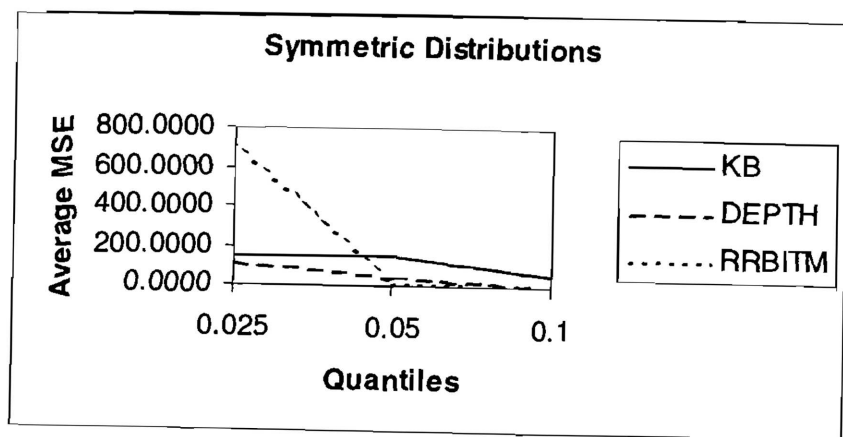


Figure 3. Average MSE for Design Matrix M3

We now remark on the graphs for the symmetric distributions.

At the higher extreme quantile of 0.1, all estimators do reasonably well with KB having some problems at M3 (as expected). The divergence in the performance of the estimators is more pronounced at the more extreme quantiles 0.05 and 0.025. At 0.05, KB performs poorly when influential observations are present. The most interesting case is M3 where we see RRBITM's inability to handle 2 influential points at the 0.025 quantile, but dramatically improving its performance as soon as one moves to the inner extreme quantiles.

Consider now the gamma distributions. We do not show the graphs of the lower quantiles since these are in the uninteresting short tail of the distributions. For the upper quantiles only design M4 has an influential point affecting these quantiles. M1, M2, M3 and M5 are quite similar. We thus only give the upper quantile graphs for M1 and M4.

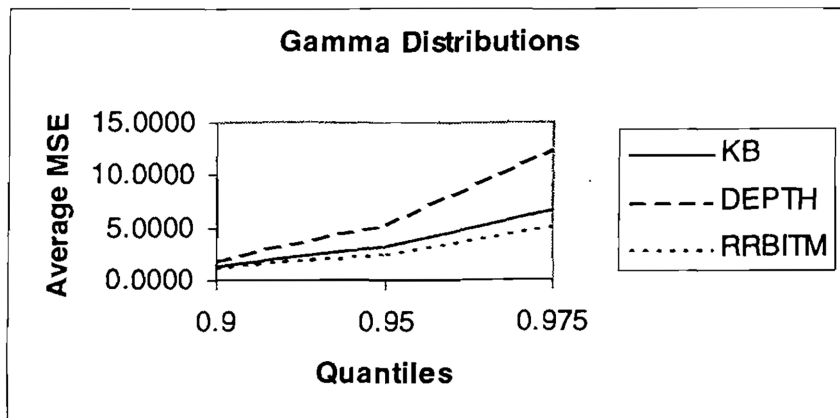


Figure 4. Average MSE for Design Matrix M1

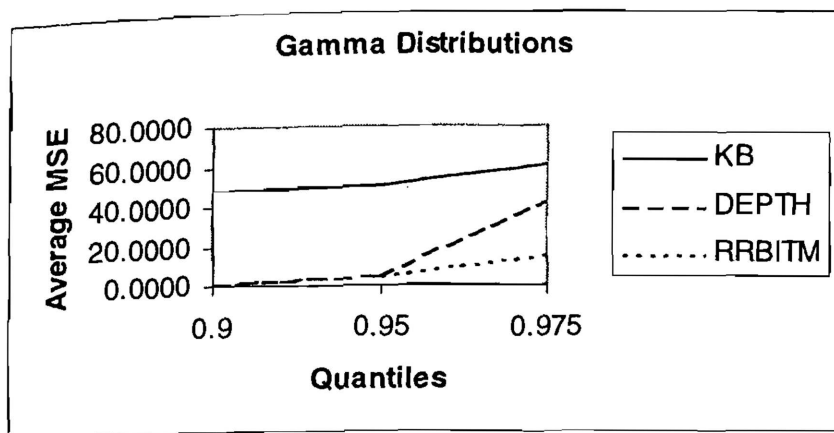


Figure 5. Average MSE for Design Matrix M4

These graphs do not have any unexpected features and the restricted, bounded influence estimators again have superior performance.

5. Conclusions

In this paper we studied the finite sample behaviour of a range of estimators for extreme regression quantiles. This was done over a range of design matrices and a range of error distributions. Our main criterion of performance was the MSE of estimation. Although we studied both the slope and intercept estimators, our focus was primarily on the latter since this is the more important parameter in extreme value studies.

Our main conclusion is that restricted bounded influence estimators should be the estimators of choice. The main problem of the nonrestricted ones is a bias problem. In analysing MSE in terms of its variance and bias components, it was seen that in this case, the MSE is usually inflated because of inflated bias (rather than inflated variance). Within the class of restricted bounded influence estimators, the Koenker-Bassett ones are preferred to the depth ones, although

their performance is very similar. The Koenker-Bassett estimators have a major advantage of computational simplicity, also for $p > 2$. Within the Koenker-Bassett class, RRBITM performed slightly better than RRBIBK. Furthermore, also for the slope parameter, the bounded influence estimators performed very well in all cases with small loss of efficiency at well-behaved designs. The results and conclusions were given for the case of $n = 41$. Similar conclusions hold for $n = 21$.

In conclusion, in estimating extreme regression quantiles in the potential presence of heavy tailed error distributions and influential design points, the preferred estimator is a restricted bounded influence Koenker-Bassett based on a common slope estimator of a 20% trimmed mean. This estimator has good performance in well-behaved cases and protects one against the adverse effects of heavy tails and high leverage design points.

Note, however, as suggested by one of the referees, that this good performance holds for the pure location shift model. If serious heteroscedasticity exists, there would be a real bias-variance trade-off. We will return to this issue in our further research.

Acknowledgement

The authors would like to thank the referees for their comments which substantially improved the readability of the paper.

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